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IMSciences Peshawar

Institute of Management Sciences
04 NOV 2024
Examination

Exam	:	**Mid-Term**
Program	:	**BCS Session 2022-26**
Department	:	**Computer Science**
Paper	:	**Computer Networks**

Best

Time allowed: 20 minutes **Total Marks: 30**

Name: Muhammad Shahan

SECTION – A **Marks: 15**

Note: Section-A is compulsory. It should be completed in the first 15 minutes and handed over to the teacher. Cutting/overwriting is not allowed. Do not use lead pencil.

14/15

Q1. Attempt the following MCQs.

1. Which of the following is an example of Bluetooth?
a) wide area network
b) virtual private network
c) local area network
(d) personal area network

2. What are nodes in a computer network?
a) the computer that routes the data
b) the computer that terminates the data
c) the computer that originates the data
(d) all of the mentioned

3. Which one of the following is not a function of network layer?
(a) congestion control b) error control c) routing d) inter-networking

4. How is a single channel shared by multiple signals in a computer network?
(a) multiplexing b) phase modulation c) analog modulation d) digital modulation

5. The port numbers, along with the source and destination addresses for the data, determine
a) port b) machine (c) socket d) pipe

7. Which layer does the data link layer take packets from and encapsulate them into frames for transmission?
a) transport layer b) application layer (c) network layer d) physical layer

8. Which type of network shares the communication channel among all the machines?
a) anycast network b) multicast network c) unicast network (d) broadcast network

9. Which topology requires a multipoint connection?
a) Ring (b) Bus c) Star d) Mesh

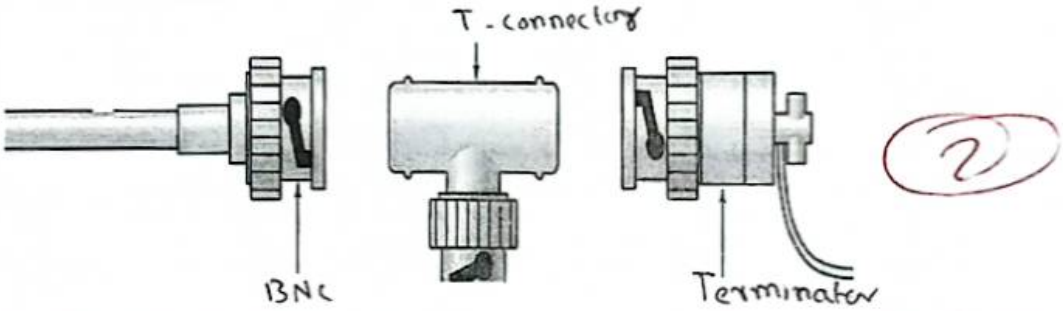
10. What was the name of the first network?
a) ASAPNET (b) ARPANET c) CNNET d) NSFNET

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11. Name the following connectors



12. Which type of unguided media requires the transmitter and receiver to be aligned due to its unidirectional nature?

a) Radio waves b) Infrared (c) Microwave d) Optical Fiber

13. Which propagation method allows radio waves to travel long distances by reflecting off the ionosphere?

a) Ground Propagation
b) Line-of-Sight Propagation
c) Space Propagation
(c) Sky Propagation

14. What advantage does Unshielded Twisted Pair (UTP) cable offer over Shielded Twisted Pair (STP)?

a) Higher bandwidth
b) Immunity to electromagnetic interference
(c) Lower cost
d) Greater durability

15. Which type of fiber-optic cable reduces signal distortion by using varying densities within the core?

a) Multimode Step-Index Fiber
(b) Multimode Graded-Index Fiber
c) Single Mode Fiber
d) Coaxial Fiber

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Best

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INSTITUTE OF MANAGEMENT SCIENCES | PESHAWAR

Name: Muhammad Shahan Ibad **Registration No:** **Serial No:** 36149

Name of the Programme: BS DS - **Session:** **Semester:** 4TH **Group:**

Name of the Course: CN - **Examination:** **Date:**

Serial No. of continuation sheets, if received.

Q. No.	Total Marks	Marks Obtained
I	15	14
II	3	3
III	5	5
IV	7	7
V		
VI		
VII		
VIII		
IX		
X		
Total:	30	29

Examiner's Full Signature

START WRITING FROM THE NEXT PAGE

Examination
Examination Seal

INSTRUCTIONS

1	Students will be allowed to enter the examination room for upto 20 minutes after the start of the examination. No extra time shall be given to latecomers for any reason.
2	Students are allowed to leave if they finish their exam early, but not during the first and last 20 minutes of the examination.
3	No student will be allowed to appear in an examination without Clearance Slip (attendance, finance and QEC) and student ID Card; ID Cards, Clearance Slip must be kept displayed during the examination.
4	Each student is required to bring his/her own calculator (wherever allowed), ruler, pen during the examination; borrowing of these items are strictly prohibited.
5	Writing anything on the question paper, except examination details, is prohibited.
6	All the helping gadgets such digital dairy, notebook/laptop, mobile phone, smart watch, ear buds, etc. will be strictly prohibited in the examination room. Therefore, students are instructed to avoid bringing all such items in the examination room.
7	Students / candidates are not allowed to talk to each other inside examination hall/ room.
8	Students must read the question paper carefully and make sure to follow the instruction given therein.

Student's Signature: _____

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QUESTION FIRST

RING VS MESH TOPOLOGY

* **Mesh Topology:-**

Topology:- The Physical and logical arrangment of network is called topology.

We have different kind of topology, here we will highlight Rmg and Mesh topology.

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Definition With Explanation:- In Mesh Topology every device has a dedicated link with every other device.

Mesh Topology is very complicated because of many wires. If a device wants to send the data with intented device They can send it through own dedicated link or also have options of extra link.

Types:- In Mesh we have two topology.

2 TRUE MESH:- True Mesh has no extra link and all devices are fully connected.

HYBRID MESH:- Hybrid Mesh has a extra link but not fully connected.

ADVANTAGES:-

(*) Mesh Topology is Very reliable.

(*) Mesh is also robust if one link is fail, entire network won't down

(*) In Mesh every device is point to point connection sending and recieving will easy.



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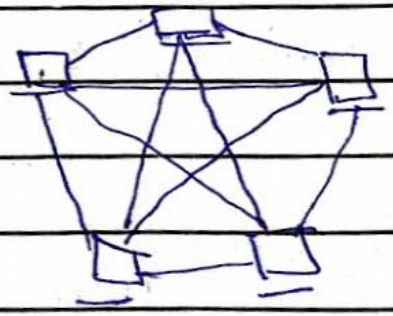
DISADVANTAGE:-

(*) Required more Number of ports and I/O.

(*) Installation is difficult ✓

(*) Alot of wire required to connect devices. ✓

DIAGRAM:-



Mesh topology -

RING TOPOLOGY:- In Ring Topology, every device has a dedicated link on either side of it.

Devices are connected in such away it making a ring shape.

If a device wants to send a Data it will Circulate in a ring and intented device will recive it by seeing it address and other will discard. After recieving it will send aeknowledgment signal also

Example:-

FDDI is a best example of Ring topology.

Marks

Total:

Advantages:-

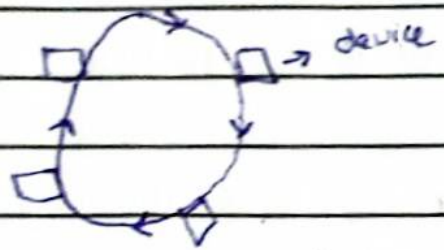
- it is good for transmission because adjacently connected
- we can add more device easily
- it broadcast the data.

DisAdvantage:-

- Difficult to troubleshoot
- If one device fail Entire network will goes down.

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DIAGRAM:-



It will circulate clock-wise.

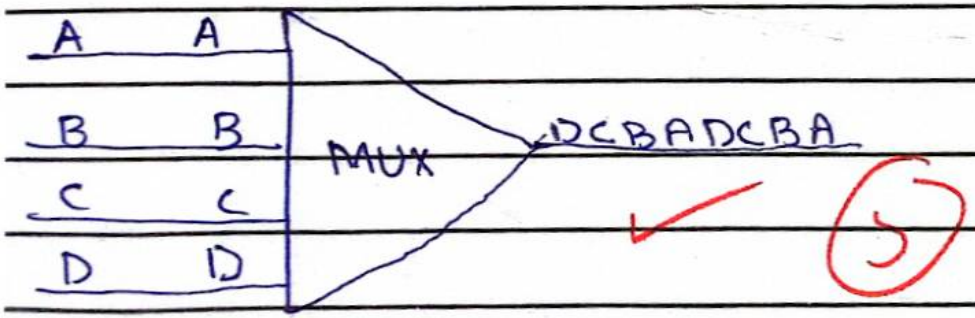
QUESTION SECOND :-

STATISTICAL TDM VS SYNCHRONOUS TDM-

TDM :- Time Division multiplexing-

SYNCHROUNS Synchronous

~~STATISTICAL~~ :- In ~~Statistical~~ TDM, every signal (channel) have fixed Time slot To send The data.



(5)

SYNCHRONOUS TDM.

Synchronous
~~Statistical~~ has fixed Time slot so everyone sends it data on that timeslot.

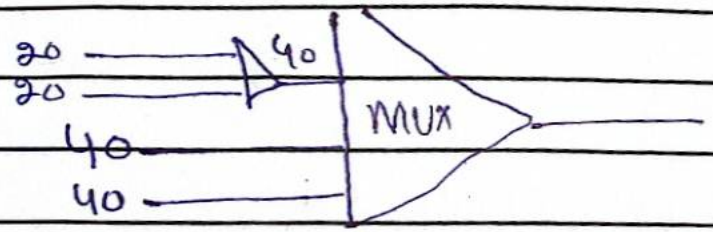
Synchronous
ISSUES WITH STASTICAL TDM.

EMPTY SLOT :- if a ~~data~~ (Channel) has no data to send. But it has own timeslot and it will the waste for it.

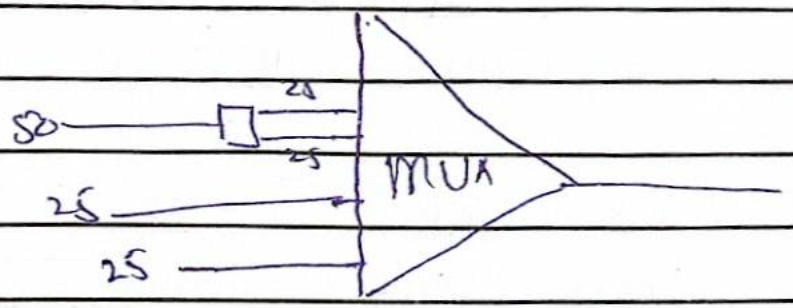
DATA RATE :-

Multilevel Multiplexing
Multiple Allocation
Pulse STUffing.

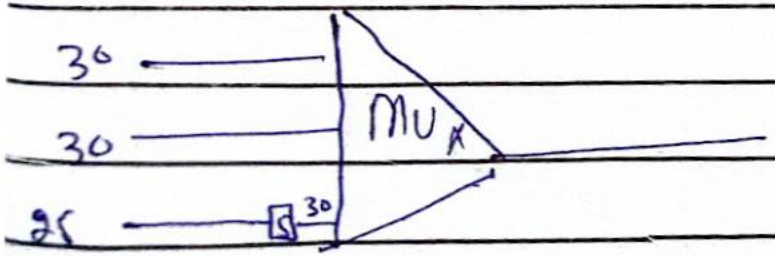
let's understand these Data rates with help of Diagram.



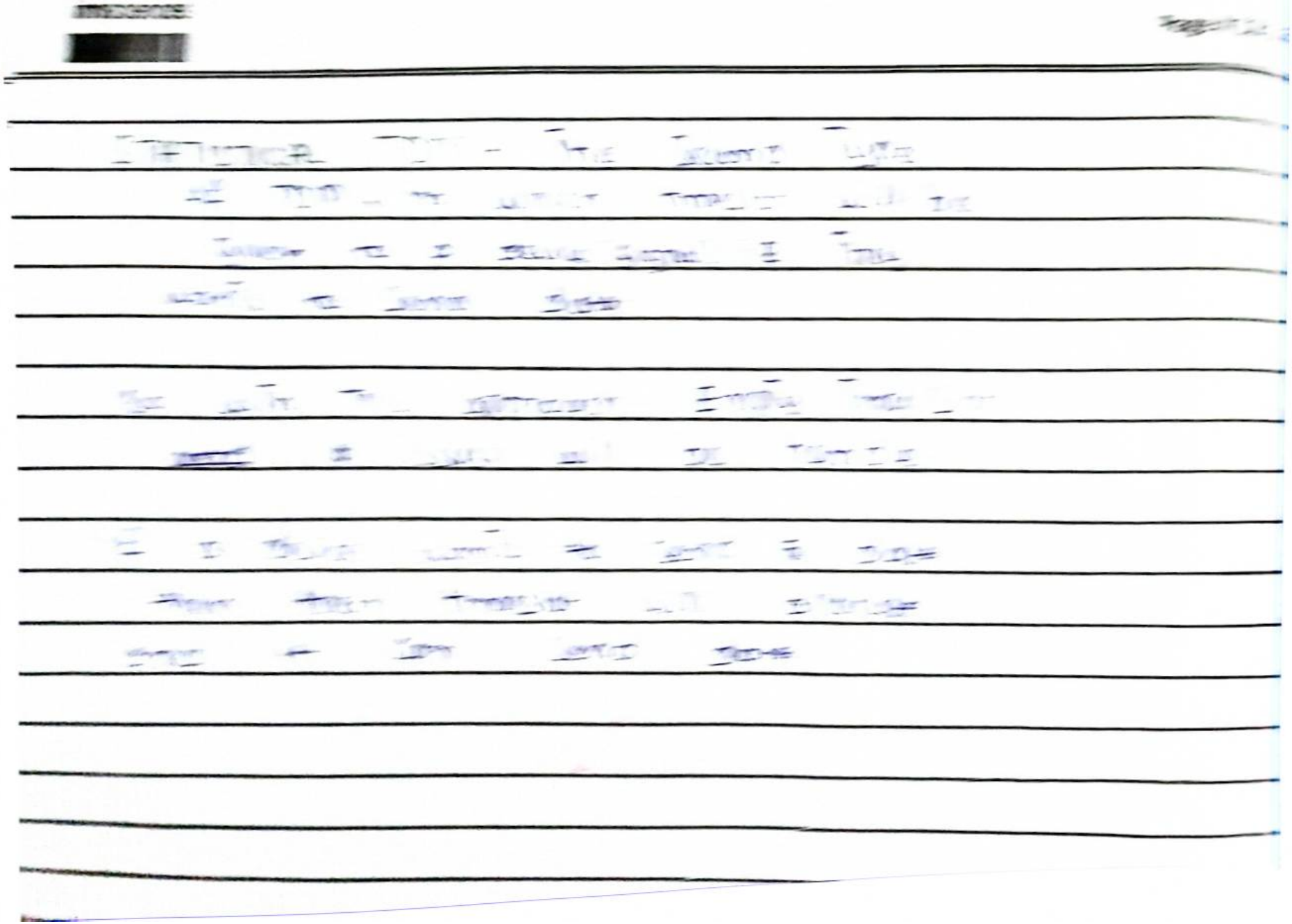
MULTI-LEVEL



MULTIPLE ALLOCATION.



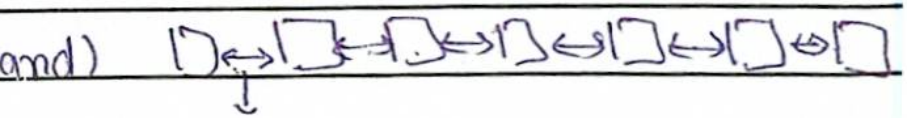
PULSE STUFFING



QUESTION THIRD

Seven channel each with 200 KHz.

for Seven channel There will be 6 (Guard band) with each of 20 KHz.



find Minimum bandwith

SOLUTION:-

$7 \times 200 + 6 \times 20 = 1520$ -

200	20
7	6
1400	120
120	
1520	

1520 KHz.